ONMATHS · ISABELLA · YEAR 8 SET 2

Simple & Compound Interest

Using FDP, multipliers and the four operations

Mr Ali · 60-minute online lesson

DO NOW · 5 MINUTES · NO CALCULATOR

Four operations warm-up

1) 250×0.04

2) 1.05×1.05

3) $3600 \div 100 \times 7$

4) 480×1.1

FDP RECAP · 5 MINUTES

Percentages → decimals → multipliers

Percentage	Fraction	Decimal	Increase by it
4%	4/100	0.04	× 1.04
15%	15/100	0.15	× 1.15
2.5%	?	?	?
30%	?	?	?

Rule: ÷ 100 to get the decimal. Add it to 1 to get the increase multiplier.

KEY WORDS

What is interest?

Principal

The money you start with (save or borrow)

Interest

Extra money the bank pays you, or you pay the bank

Rate (per annum)

The percentage added each year

Simple vs compound

Simple: same interest every year. Compound: interest on the interest too.

I DO · SIMPLE INTEREST · 3 MINUTES

£800 saved at 4% simple interest for 3 years

Step 1 · one year

800 × 0.04 = £32

Step 2 · all years

32 × 3 = £96

Step 3 · total

800 + 96 = £896

WORKING OUT

WE DO · SIMPLE INTEREST · 3 MINUTES

£1500 at 6% simple interest for 4 years. Find the total.

Step 1 · one year

1500 × ___ = ___

Step 2 · all years

___ × 4 = ___

Step 3 · total

1500 + ___ = ___

WORKING OUT

YOU DO · 4 MINUTES

Simple interest

a) £500 at 3% for 4 years. Find the **interest**.

b) £1200 at 2.5% for 2 years. Find the **total**.

c) £2000 at 5% for 6 years. Find the **total**.

WORKING OUT

COMPOUND INTEREST · THE IDEA

£1000 at 10% for 3 years

Year	Simple (always +£100)	Compound (+10% of new amount)
Start	£1000	£1000
1	£1100	£1100
2	£1200	£1210
3	£1300	£1331

Compound earns **interest on the interest**, so it grows faster.

THE SHORTCUT

$\text{Total} = \text{amount} \times \text{multiplier}^{n}$

n = number of years

$1000 \times 1.1 \times 1.1 \times 1.1 = 1000 \times 1.1^{3} = £1331$

On a calculator: 1000 × 1.1 x^{y} 3 =

MINI WHITEBOARDS · 5 MINUTES · SHOW ME!

Six quick questions

1) Multiplier for an 8% increase?

2) Multiplier for a 12% decrease?

3) £200 at 5% simple for 2 years. Interest?

4) Work out 1.2^2

5) £1000 × 1.1^2. Total?

6) 5% simple or 5% compound for 3 years — which gives more?

I DO · COMPOUND INTEREST · 3 MINUTES

£2500 at 3% compound interest for 4 years

Step 1 · multiplier

3% → 1.03

Step 2 · power

2500×1.03^4

Step 3 · round

= £2813.77

WORKING OUT

WE DO · COMPOUND INTEREST · 3 MINUTES

£3200 at 4% compound interest for 3 years

Step 1 · multiplier

4% → ___

Step 2 · power

3200 × ___

Step 3 · round

= £ ___

WORKING OUT

YOU DO · 6 MINUTES · CALCULATOR

Compound interest

a) £600 at 5% for 2 years. Find the total.

b) £4000 at 2% for 3 years. Find the total.

c) £1500 at 4% for 5 years. Find the **interest**.

WORKING OUT

OPTIONAL · EXAM-STYLE · IF TIME

Which bank is better?

Jade invests **£5000 for 5 years**.

Bank A
4% simple interest

Bank B
3.5% compound interest

Show working for both. By how much is the better bank better?

EXAM-STYLE · 3 MINUTES · CALCULATOR

GCSE-style question 1

Priya invests £2000 in an account paying 6% compound interest per year. After how many whole years will her savings first be worth more than £2500?

(3 marks)

WORKING OUT

EXAM-STYLE · 3 MINUTES · CALCULATOR

GCSE-style question 2

A laptop loses 20% of its value each year. After 2 years it is worth £448. Work out the price of the laptop when it was new.

(3 marks)

WORKING OUT

EXAM-STYLE · 3 MINUTES · CALCULATOR

GCSE-style question 3

Sam invests £3000. In the first year he gets 5% interest. For the next 2 years he gets 2% compound interest per year. How much does he have after 3 years?

(3 marks)

WORKING OUT

EXAM-STYLE · 3 MINUTES · NON-CALCULATOR

GCSE-style question 4

Ali says, “10% compound interest for 2 years is the same as 20% simple interest for 2 years.” Is Ali correct? Give a reason.

(2 marks)

WORKING OUT

OPTIONAL STRETCH · IF TIME

Going down: depreciation

A car costs **£12 000**. It loses **15%** of its value each year.

What is it worth after **3 years**?

Hint: a 15% decrease leaves 85%. What multiplier is that?

EXIT TICKET · 4 MINUTES

Three to finish

1) £400 at 5% simple interest for 3 years. Total?

2) What is the multiplier for a 6% increase?

3) £3000 at 2% compound interest for 2 years. Total?